Porphobilinogen

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Identity Sheet

Property	Value
Name	Porphobilinogen
Other names	PBG, 2-Aminomethyl-4-(2-carboxyethyl)-1H-pyrrole-3-propanoic acid
Chemical formula	$C_{10}H_{14}N_2O_4$
Molecular weight	226.23 g/mol
CAS number	487-90-1
PubChem CID	1021
SMILES	NCc1[nH]cc(CCC(=O)O)c1CC(=O)O
Appearance	White to off-white crystalline powder
Density	~1.4 g/cm^3 (estimated)
Melting point	Decomposes before melting (~170 °C)

Structural formula:

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HOOC-CH₂ CH₂-NH₂
| |
C──C
|| |
C NH
|
HOOC-CH₂-CH₂
```

Porphobilinogen is a pyrrole derivative bearing an aminomethyl group, a carboxymethyl (acetic acid) side chain, and a 2-carboxyethyl (propionic acid) side chain. It is a dicarboxylic acid with two ionisable carboxyl groups (pKa ~4.0 and ~4.5) and a primary amine (pKa ~9.9). At physiological urinary pH (4.5–8.0), PBG exists predominantly as a zwitterionic or anionic species. It is highly water-soluble and photosensitive — it can polymerise and darken when exposed to light and air, forming uroporphyrinogen-like pigments.

Molecules Not to Be Confused With

Molecule	Formula	MW (g/mol)	Key Difference
δ-Aminolevulinic acid (ALA)	$C_5H_9NO_3$	131.13	Precursor to PBG in heme pathway; δ-amino ketone, not a pyrrole; also elevated in acute porphyrias

Molecule	Formula	MW (g/mol)	Key Difference
Urobilinogen	$C_{33}H_{44}N_4O_6$	592.73	Bile pigment from bilirubin reduction; also reacts with Ehrlich's reagent (false positive in Watson-Schwartz test)
Uroporphyrinogen	$C_{40}H_{44}N_4O_{16}$	836.80	Tetrapyrrole formed from 4 PBG molecules; downstream in heme pathway
Indican (Indoxyl sulfate)	$C_8H_6NO_4SK$	251.30	Urinary tryptophan metabolite; may interfere with colorimetric tests

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Medical Information

Origin

Endogenous: Biosynthesis — Heme Synthesis Pathway PBG is the **second intermediate** in the heme biosynthesis pathway. It is synthesised in **all nucleated cells**, with the highest rates in **erythroid precursors** (bone marrow) and **hepatocytes** (liver).

Biosynthesis:

1. **ALA synthesis** (mitochondrial, rate-limiting): Glycine + Succinyl-CoA → δ-Aminolevulinic acid (ALA). Catalysed by **ALA synthase** (ALAS1 in liver, ALAS2 in erythroid cells). Requires pyridoxal phosphate (vitamin B_6) cofactor.
2. **PBG synthesis** (cytoplasmic): 2 ALA → Porphobilinogen + 2 H_2O. Catalysed by **ALA dehydratase** (also called porphobilinogen synthase, PBGS). Zinc-dependent enzyme.
3. PBG is then converted to hydroxymethylbilane by **PBG deaminase** (PBGD, also called hydroxymethylbilane synthase, HMBS), the enzyme deficient in acute intermittent porphyria (AIP).

Daily PBG production is linked to heme turnover: approximately **300 mg of heme** is synthesised daily, primarily for haemoglobin.

Exogenous PBG has **no exogenous dietary sources**. It is exclusively an endogenous metabolic intermediate. It is not found in foods and has no pharmaceutical or cosmetic uses.

Primary & Secondary Biological Roles

Primary role: - Heme precursor: PBG is an obligatory intermediate in the biosynthesis of heme, which is essential for haemoglobin, myoglobin, cytochromes, catalase, peroxidases, and other haemoproteins.

Secondary roles: - No independent biological function. PBG has no known receptor, signalling, or structural role. Its importance lies entirely as a metabolic intermediate. Accumulation of PBG (and ALA) is neurotoxic and is believed to be responsible for the neurological symptoms of acute porphyrias, possibly through effects on GABAergic neurotransmission and oxidative stress.

Catabolism and Elimination Pathway

- Under normal conditions, PBG is **rapidly consumed** by PBG deaminase in the heme biosynthetic pathway, with minimal accumulation.
- A small amount of PBG is **excreted in urine** by glomerular filtration (freely filtered, MW 226 Da).

- PBG is **not significantly reabsorbed** by renal tubules.
- In acute porphyria attacks, the bottleneck at PBG deaminase causes massive PBG accumulation; urinary PBG excretion can increase **10–100 fold**.
- PBG is **unstable in urine**: it spontaneously polymerises and oxidises (especially in light and at alkaline pH) to form porphyrins and other pigments, causing urine to darken on standing ("port-wine" discoloration).

Expression in Humans

Normal Levels

Compartment	Reference Range
Plasma PBG	< 0.12 µmol/L (rarely measured clinically)
Urinary PBG (24-h)	< 8.8 µmol/day (< 2 mg/day)
Urinary PBG (random, creatinine-corrected)	< 1.5 µmol/mmol creatinine
Urinary PBG (random)	< 8.8 µmol/L

Factors Influencing Levels **Increased PBG:** - Acute intermittent porphyria (AIP) — acute attacks - Variegate porphyria (VP) — acute attacks - Hereditary coproporphyria (HCP) — acute attacks - ALA dehydratase deficiency porphyria (ADP) — rare - Precipitating factors for porphyria attacks: fasting, certain drugs (barbiturates, sulfonamides, oral contraceptives), alcohol, infections, stress, menstrual cycle (progesterone)

Decreased PBG: - Not clinically significant in isolation - Very low ALA production (severe lead poisoning inhibits ALA dehydratase → ALA rises but PBG may fall) - Adequate heme supply (negative feedback on ALA synthase reduces pathway flux) - Well-nourished, unstressed state - No known pathology from PBG deficiency per se

Associated Pathologies

Condition	PBG Level	Key Symptoms
Acute intermittent porphyria (AIP)	Urinary PBG >10× ULN during attacks (>50 µmol/day, often >150 µmol/day)	Severe abdominal pain, nausea, vomiting, tachycardia, hypertension, psychiatric symptoms, motor neuropathy, seizures
Variegate porphyria (VP)	Moderately elevated PBG during neurovisceral attacks	Abdominal pain + photosensitive skin lesions
Hereditary coproporphyria (HCP)	Moderately elevated PBG during attacks	Similar to AIP + occasional photosensitivity
ALA dehydratase deficiency porphyria	Very low PBG (ALA elevated instead)	Extremely rare; abdominal pain, neuropathy
Lead poisoning	Normal or slightly elevated PBG; ALA markedly elevated	Abdominal colic, anaemia, encephalopathy, wrist drop

Presence in Urine

Should it be normally present? Yes, in **trace amounts**. PBG is a small molecule freely filtered at the glomerulus. Normal urinary PBG is very low (< 2 mg/day) and is not routinely detectable by standard dipstick tests.

Normal urinary levels: < 8.8 µmol/day (< 2 mg/day); random urine < 1.5 µmol/mmol creatinine.

Form in urine: PBG is present in its **native molecular form**, primarily as the dianionic species at typical urine pH. It is unstable and can polymerise to porphyrin-like pigments, particularly at alkaline pH and in the presence of light.

Pathological significance:

Urinary PBG	Possible Causes	Prevalence
Markedly elevated (>10× ULN)	Acute porphyria attack (AIP, VP, HCP)	AIP prevalence: ~1:20,000 (higher in certain populations, e.g. Scandinavia ~1:10,000)
Mildly elevated (2–10× ULN)	Latent porphyria carrier in remission, subclinical porphyria, certain drugs	Variable
Normal	Healthy state; also lead poisoning (ALA elevated, PBG normal)	—

Solubility: PBG is **highly soluble** in water (>100 g/L). It does not form crystals or stones. Its instability (polymerisation) is the main concern, not insolubility.

Detection in Urine

Available Clinical Assays

1. **Quantitative ion-exchange chromatography + Ehrlich's reagent (gold standard):**
   - **Principle:** Urine is passed through an anion-exchange resin column (e.g., Bio-Rad 1-X2) that retains PBG while washing away interfering substances (urobilinogen, indican). PBG is eluted and reacted with Ehrlich's reagent (p-dimethylaminobenzaldehyde in HCl) to form a red-pink aldehyde adduct (Ehrlich chromogen).
   - **Detection:** Absorbance at **553 nm** (some protocols: 555 nm).
   - **LOD:** ~0.5 µmol/L (~0.1 mg/L).
   - **Advantages:** Specific, quantitative, removes interfering chromogens.
   - **Disadvantages:** Multi-step, requires column, manual labour, ~30 min.
2. **Watson-Schwartz test (qualitative screening):**
   - **Principle:** Urine mixed with Ehrlich's reagent; colour extracted with chloroform and butanol to distinguish PBG from urobilinogen (urobilinogen partitions into chloroform; PBG stays in aqueous phase).
   - **Detection:** Visual colour assessment — pink-red in aqueous layer = positive for PBG.
   - **LOD:** ~50 µmol/L (~11 mg/L) — insensitive, misses mild elevations.
   - **Advantages:** Rapid (<5 min), bedside.
   - **Disadvantages:** Semi-quantitative, poor sensitivity, false positives from drugs.
3. **Hoesch test (qualitative screening):**
   - **Principle:** Two drops of urine added to Ehrlich's reagent (reverse addition vs Watson-Schwartz). Immediate cherry-red colour = PBG positive.
   - **Detection:** Visual colour.
   - **LOD:** ~25–50 µmol/L.
   - **Advantages:** Faster and simpler than Watson-Schwartz; fewer false positives.
   - **Disadvantages:** Still semi-quantitative.
4. **LC-MS/MS (research/reference labs):**
   - **Principle:** Liquid chromatography with tandem mass spectrometry for direct quantification.
   - **Detection:** Mass-to-charge ratio; highly specific.
   - **LOD:** ~0.01–0.1 µmol/L.
   - **Advantages:** Gold-standard specificity, multiplexed with ALA.
   - **Disadvantages:** Expensive instrumentation, not widely available.

Optimal Urine Type for Measurement

- **Random (spot) urine** is preferred for emergency screening of suspected acute porphyria attacks — results needed urgently.
- **24-hour urine** for quantitative confirmation and monitoring.
- PBG/creatinine ratio on a random sample correlates well with 24-h excretion.
- **Protect sample from light** (wrap in foil) and keep at **2–8 °C** or freeze at **–20 °C**. PBG is stable in the dark at 4 °C for ~48 h and at –20 °C for ≥1 month.
- **Do not alkalinise** the urine (accelerates PBG polymerisation).
- First morning urine offers no particular advantage; timing should not delay testing if acute attack is suspected.

Actual Gold Standard

The **quantitative ion-exchange column method with Ehrlich's reagent** (Mauzerall-Granick method or Bio-Rad PBG kit) remains the primary reference method in most clinical laboratories. It is performed on the Beckman DU, Shimadzu, or similar spectrophotometers at 553 nm. CV is typically 5–10%. LC-MS/MS methods are increasingly used in reference laboratories.

Interferences in Measurement

Interference	Effect	Mechanism
Urobilinogen	False positive (Watson-Schwartz)	Also reacts with Ehrlich's reagent; removed by chloroform extraction or ion-exchange
Indican / Indoxyl sulfate	False positive	Ehrlich's reagent cross-reactivity
Methyldopa, phenazopyridine	False positive (colour)	Produce coloured compounds in acidic conditions
Light exposure	False decrease	PBG polymerises to porphyrins (lost from PBG measurement)
Alkaline pH	False decrease	Accelerates PBG degradation
Bacterial contamination	Variable	May degrade PBG
High urobilinogen	Interference with direct Ehrlich methods	Overcome by column chromatography step

Research Detection Methods

Spectroscopy Detection (UV-Vis / NIR)

- **UV-Vis:** PBG itself absorbs weakly in the UV range. The pyrrole ring shows absorption around **220–240 nm** ($\pi \rightarrow \pi^*$), but this is not specific in a urine matrix. The primary UV-Vis approach is indirect via the **Ehrlich chromogen** (PBG–p-DMAB adduct), which absorbs strongly at **553 nm** ($\varepsilon \approx 6.1 \times 10^4\ M^{-1}cm^{-1}$).
- **LOD (Ehrlich-based spectrophotometry):** ~0.5 µmol/L after ion-exchange cleanup.
- **Sample prep:** Column chromatography (anion exchange) to isolate PBG from urine matrix, then Ehrlich's reagent addition.
- **NIR:** No specific NIR methods have been reported for PBG quantification in urine. PBG concentrations in normal urine are far below typical NIR detection limits.

Fluorescence Detection

- PBG itself shows **weak native fluorescence** from the pyrrole ring (Ex ~280 nm / Em ~340 nm), but this is insufficient for direct detection in urine.

- **Fluorescence spectrophotometry of porphyrin products:** A custom device for two-step screening measures PBG by converting it to fluorescent porphyrins: PBG is first condensed to uroporphyrinogen (using PBG deaminase), then oxidised to uroporphyrin. Uroporphyrin fluorescence is measured at **Ex 405 nm / Em 596 and 620 nm** (Soret excitation, Q-band emission).
  - LOD: ~2 μmol/L for PBG (Lang et al., 2015, JBO; doi:10.1117/12.2183855).
  - Sample prep: Addition of PBG deaminase enzyme + oxidant; 30 min incubation.
- **Spectrofluorimetric methods (Hantzsch reaction):** PBG reacted with acetylacetone + formaldehyde to form a fluorescent derivative. Derivative matrix-isopotential synchronous spectrofluorimetry allows simultaneous ALA and PBG determination.
  - Ex/Em: Hantzsch product fluorescence in the range **Ex 360–410 nm / Em 470–510 nm**.
  - LOD: ~0.1 μmol/L (Frontiers in Chemistry, 2022).
  - Sample prep: Chemical derivatisation + heating.

Raman Detection No dedicated Raman studies for PBG quantification in urine have been published. PBG's pyrrole ring would show expected Raman features:

Peak (cm^{-1})	Assignment (predicted)
~1370–1400	Pyrrole ring C=C/C–N stretches
~1550–1580	Pyrrole ring C=C stretch
~1000–1050	C–C stretch of side chains
~2900–3000	C–H stretches

- At normal urinary concentrations (<10 μmol/L), conventional Raman detection is not feasible. SERS might potentially detect PBG but no published studies exist.
- **Research gap:** No LOD, excitation, or sample prep data available for Raman/SERS of PBG.

FTIR Detection No dedicated FTIR methods for PBG in urine have been published. Expected infrared bands of PBG:

Band (cm^{-1})	Assignment	Notes
~3300–3500	N–H stretch (pyrrole NH + primary amine)	Broad, overlaps with water
~1700–1720	C=O stretch (carboxylic acids)	Two COOH groups
~1580–1600	NH_2 scissoring / COO^- asymmetric stretch	If ionised
~1400–1450	COO^- symmetric stretch	—
~1350–1400	Pyrrole ring modes	—

- PBG is present in trace amounts in normal urine, well below FTIR detection limits for minor components (~1 mM minimum for ATR-FTIR).
- Even during acute attacks (PBG ~50–500 μmol/L), concentrations remain below typical ATR-FTIR sensitivity without preconcentration.
- **Research gap:** No published LOD or method for FTIR detection of urinary PBG.

Voltammetry Detection

- PBG contains an electrochemically active pyrrole ring and primary amine, which can be oxidised.

- **Direct oxidation:** The pyrrole moiety can be oxidised at carbon electrodes at potentials around **+0.8 to +1.0 V vs Ag/AgCl** in acidic media. However, no validated analytical methods for urinary PBG by voltammetry have been published.
- **Biosensor concepts:** Theoretically, a PBG deaminase-based biosensor could detect PBG consumption, but no such sensor has been reported.
- **Research gap:** No LOD, selectivity, or real-urine validation data available for electrochemical PBG detection.

Other Detection Technologies

1. **HPLC with UV detection:**
   - Ion-pair HPLC on C18 columns with UV detection at 240 nm.
   - LOD: ~0.1–0.5 µmol/L.
   - Sample prep: Acidification, centrifugation, direct injection.
   - Reference: Jamani et al., Clin Chem 1989.
2. **Rapid dipstick / point-of-care tests:**
   - Experimental paper-based strips using Ehrlich's reagent impregnated on filter paper.
   - LOD: ~25–50 µmol/L (similar to Hoesch test).
   - Advantage: Bedside, no equipment. Limitation: Semi-quantitative.
3. **Capillary electrophoresis:**
   - Separation and UV detection of PBG in urine.
   - LOD: ~0.5 µmol/L.
   - Sample prep: Dilution, filtration.

Summary Table: Detection Methods Comparison

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Ion-exchange + Ehrlich's (gold standard)	~0.5 µmol/L	553 nm (Ehrlich chromogen)	Column chromatography + reagent	Quantitative, specific	Labour-intensive, 30 min
Watson-Schwartz test	~50 µmol/L	Visual pink-red	Ehrlich's + chloroform extraction	Rapid bedside screening	Poor sensitivity, false positives
Hoesch test	~25–50 µmol/L	Visual cherry-red	Reverse Ehrlich's addition	Simpler than Watson-Schwartz	Semi-quantitative
LC-MS/MS	~0.01–0.1 µmol/L	m/z specific	Dilute and shoot or SPE	Highest specificity, multiplex	Expensive, specialised
Fluorescence (porphyrin conversion)	~2 µmol/L	Ex 405 / Em 596,620 nm	Enzyme conversion + oxidation	Non-destructive, rapid potential	Requires PBG deaminase enzyme
Fluorescence (Hantzsch)	~0.1 µmol/L	Ex 360–410 / Em 470–510 nm	Derivatisation + heating	Sensitive, simultaneous ALA+PBG	Chemical derivatisation needed
HPLC-UV	~0.1–0.5 µmol/L	240 nm	Acidify, centrifuge, inject	Quantitative, automated	HPLC instrumentation

Method	LOD	Wavelength / Key Parameter	Sample Prep	Strengths	Limitations
Raman / SERS	Not established	—	—	—	No published methods for PBG
FTIR	Not established	—	—	—	Concentrations too low
Voltammetry	Not established	~+0.8–1.0 V vs Ag/AgCl (predicted)	—	—	No validated methods

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